

Exp:1A

Helloworld.java

```
import java.applet.*;
```

```
import java.awt.*;
```

```
public class HelloApplet extends Applet {
```

```
    public void paint(Graphics g) {
```

```
        g.drawString("Hello World", 50, 50);
```

```
    }
```

```
}
```

Helloworld.html

```
<html>
```

```
<body>
```

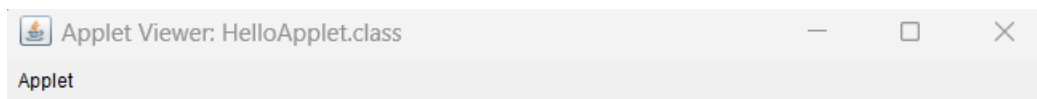
```
<applet code="HelloApplet.class" width="300" height="200">
```

```
</applet>
```

```
</body>
```

```
</html>
```

Output



Hello World

Applet started.

Exp:1B

```
import javax.swing.*;  
import java.awt.*;  
import java.awt.event.*;
```

```
public class KeyEventDemo extends JFrame implements KeyListener {
```

```
    String msg = "Assign1: A JAVA program to demo status of key";
```

```
    public KeyEventDemo() {  
        setTitle("Key Event Demo");  
        setSize(400, 200);  
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);  
        addKeyListener(this);  
        setVisible(true);  
    }
```

```
    public void keyReleased(KeyEvent k) {  
        msg = "Key Released";  
        repaint();  
    }
```

```
    public void keyTyped(KeyEvent k) {  
        msg = "Key Typed";  
        repaint();  
    }
```

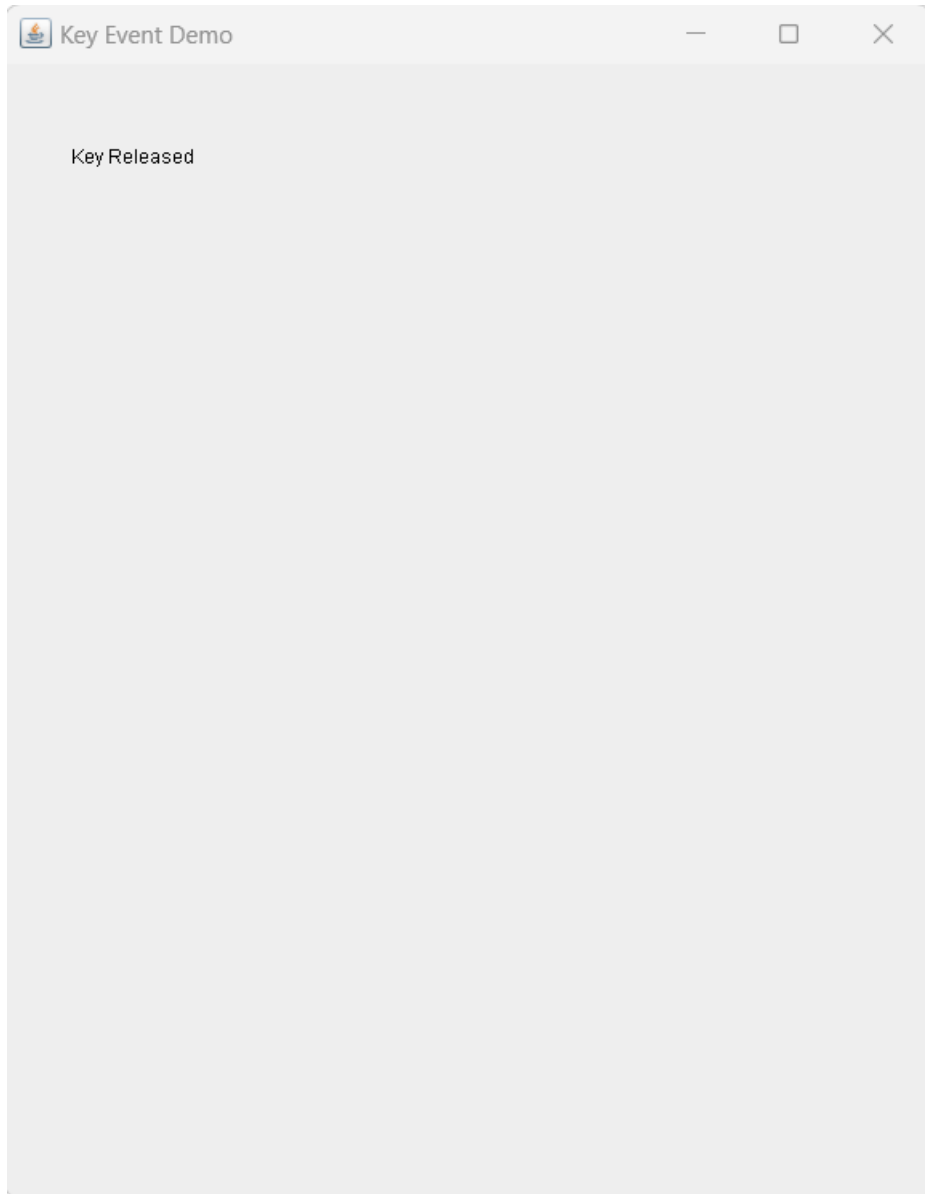
```
    public void keyPressed(KeyEvent k) {  
        msg = "Key Pressed";
```

```
        repaint();
    }

    public void paint(Graphics g) {
        super.paint(g);
        g.drawString(msg, 50, 100);
    }

    public static void main(String[] args) {
        new KeyEventDemo();
    }
}
```

Output



Exp2

```
import java.awt.*;
import java.awt.event.*;

public class MouseFrameDemo extends Frame implements MouseListener {

    String msg = "Move mouse into the frame";

    public MouseFrameDemo() {
        setTitle("Mouse Event Demo");
        setSize(400, 300);
        addMouseListener(this);
        setVisible(true);
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent we) {
                System.exit(0);
            }
        });
    }

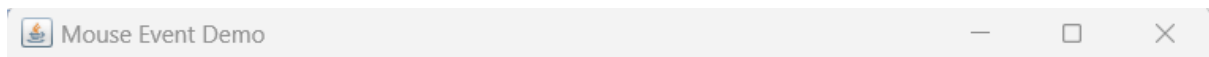
    public void mouseClicked(MouseEvent e) {
        msg = "Mouse Clicked";
        repaint();
    }

    public void mouseEntered(MouseEvent e) {
        msg = "Mouse Entered";
        repaint();
    }

    public void mouseExited(MouseEvent e) {
        msg = "Mouse Exited";
    }
}
```

```
        repaint();
    }
    public void mousePressed(MouseEvent e) {}
    public void mouseReleased(MouseEvent e) {}
    public void paint(Graphics g) {
        g.drawString(msg, 100, 150);
    }
    public static void main(String[] args) {
        new MouseFrameDemo();
    }
}
```

Output



Mouse Exited

Exp 3

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class StudentResultGUI extends Frame implements ActionListener {
```

```
    Label l1, l2, l3, l4, l5;
```

```
    TextField t1, t2, t3, t4, t5;
```

```
    Button btn;
```

```
    public StudentResultGUI() {
```

```
        setTitle("Enter Your Marks Out of 100");
```

```
        setSize(400, 300);
```

```
        setLayout(new GridLayout(6, 2));
```

```
        l1 = new Label("MIC:");
```

```
        l2 = new Label("DBMS:");
```

```
        l3 = new Label("FJP:");
```

```
        l4 = new Label("DC:");
```

```
        l5 = new Label("EFT");
```

```
        t1 = new TextField();
```

```
        t2 = new TextField();
```

```
        t3 = new TextField();
```

```
        t4 = new TextField();
```

```
        t5 = new TextField();
```

```
        btn = new Button("Show Result");
```

```
        btn.addActionListener(this);
```

```
        add(l1); add(t1);
```

```
        add(l2); add(t2);
```

```
        add(l3); add(t3);
```

```
        add(l4); add(t4);
```



```
add(l5); add(t5);
```

```
add(btn);
```

```
addWindowListener(new WindowAdapter() {  
    public void windowClosing(WindowEvent we) {  
        System.exit(0);  
    }  
});
```

```
setVisible(true);  
}
```

```
public void actionPerformed(ActionEvent e) {  
    int m1 = Integer.parseInt(t1.getText());  
    int m2 = Integer.parseInt(t2.getText());  
    int m3 = Integer.parseInt(t3.getText());  
    int m4 = Integer.parseInt(t4.getText());  
    int m5 = Integer.parseInt(t5.getText());
```

```
    int total = m1 + m2 + m3 + m4 + m5;
```

```
    double percentage = total / 5.0;
```

```
    String result;
```

```
    if (percentage >= 40)
```

```
        result = "PASS";
```

```
    else
```

```
        result = "FAIL";
```

```
    new ResultFrame(total, percentage, result);
```

```
}
```

```
public static void main(String[] args) {  
    new StudentResultGUI();  
}  
}  
  
class ResultFrame extends Frame {  
  
    public ResultFrame(int total, double percentage, String result) {  
        setTitle("Result");  
        setSize(300, 200);  
        setLayout(new GridLayout(3, 1));  
  
        add(new Label("Total Marks: " + total));  
        add(new Label("Percentage: " + percentage));  
        add(new Label("Result: " + result));  
  
        setVisible(true);  
  
        addWindowListener(new WindowAdapter() {  
            public void windowClosing(WindowEvent we) {  
                dispose();  
            }  
        });  
    }  
}
```

Output

Enter Your Marks Out of 100

MIC:

35

DBMS:

90

FJP:

95

DC:

40

EFT

28

Show Result

Result

Total Marks: 268

Percentage: 57.6

Result: PASS

Exp 4B

```
import javax.swing.*;
```

```
import java.awt.event.*;
```

```
public class MenuDemo extends JFrame implements ActionListener {
```

```
    JMenuBar mb;
```

```
    JMenu file, edit, view;
```

```
    JMenuItem newItem, openItem;
```

```
    public MenuDemo() {
```

```
        setTitle("Menu Bar Example");
```

```
        setSize(400, 300);
```

```
        // Create Menu Bar
```

```
        mb = new JMenuBar();
```

```
        // Create Menus
```

```
        file = new JMenu("File");
```

```
        edit = new JMenu("Edit");
```

```
        view = new JMenu("View");
```

```
        // Create Submenu Items under File
```

```
        newItem = new JMenuItem("New");
```

```
        openItem = new JMenuItem("Open");
```

```
        // Add action listeners
```

```
        newItem.addActionListener(this);
```

```
        openItem.addActionListener(this);
```

```

// Add submenu items to File menu
file.add(newItem);
file.add(openItem);

// Add menus to MenuBar
mb.add(file);
mb.add(edit);
mb.add(view);

// Set MenuBar to Frame
setJMenuBar(mb);

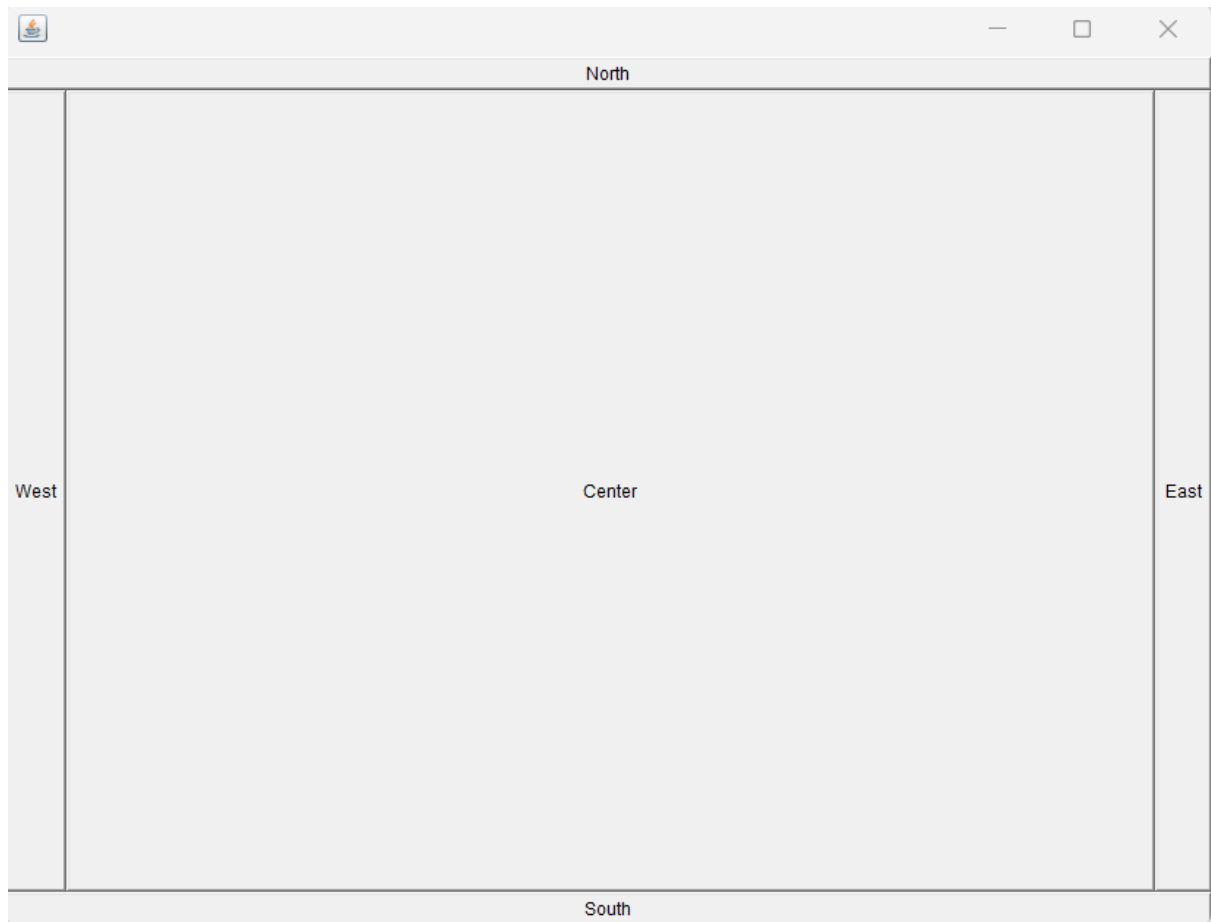
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setVisible(true);
}

// Handle Menu Clicks
public void actionPerformed(ActionEvent e) {
    if (e.getSource() == newItem) {
        JOptionPane.showMessageDialog(this, "New File Selected");
    } else if (e.getSource() == openItem) {
        JOptionPane.showMessageDialog(this, "Open File Selected");
    }
}

public static void main(String[] args) {
    new MenuDemo();
}
}

```

Output



Output

